

SugamSetu (AccessChain) — SIH 2026 Master Guide

TECH STACK CHEATSHEET & 15 CRITICAL JUDGE Q&A; WITH PRECISE WINNING ANSWERS

Core Pitch Thesis: *'Don't just find an accessible destination. Complete an accessible journey.'* Traditional platforms evaluate isolated pins. SugamSetu models transit, venue gates, elevators, corridors, and seating as a continuous **Accessibility Continuity Graph** with multi-constraint matching, SPOF detection, and real-time outage rerouting.

1. Technical Stack — Simpler, Brief, Precise & Clear

Layer	Technologies Used	Why This Tech? (Simple & Precise Explanation for Judges)
Frontend & UI	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Leaflet Maps	Delivers sub-second server rendering, WCAG 2.1 AAA accessible UI (high-contrast, voice screen-reader optimizations), and step-by-step turn navigation.
Core Routing Engine	Dijkstra / A* Multi-Constraint Pathfinding, Zustand State Machine	Evaluates entire journeys as connected node-edges. Filters infeasible links by disability profile (slope/lift) and detects Single Points of Failure (SPOFs).
Database & Spatial	PostgreSQL, Prisma ORM, PostGIS Spatial Extensions	Performs sub-50ms geospatial queries and enforces Tri-State Logic (UNKNOWN != FALSE) —guaranteeing unverified routes are never falsely assumed safe.
Multimodal AI	Gemini Vision API, Vernacular NLP (English + 5 Indic Languages)	1) Scans photos of ramps/stairs to verify obstacle compliance; 2) Provides voice-activated conversational trip assistance in regional languages.
Telemetry & Live Rerouting	MQTT / WebSockets Telemetry, Dynamic Graph Edge Invalidation	Ingests live sensor/operator reports (e.g. lift outage at Gate 3) and automatically triggers 1-click fallback rerouting (via Gate 5 ramp) in real time.

2. 15 Likely SIH Judge Questions & Exact Winning Answers

Q1: How is this different from Google Maps, Wheelmap, or Sugamya Bharat App?	[Domain & Differentiation]
Answer: Google Maps routes for speed and ignores internal barriers. Wheelmap only shows static, isolated pins (e.g. 'accessible cafe'), but ignores transit. If an accessible taxi or venue lift breaks, the trip collapses. SugamSetu models end-to-end continuous journey graphs —validating every transit, gate, elevator, and corridor link before you leave.	
Q2: What is your core mathematical/algorithmic innovation?	[Algorithms & Graph Theory]
Answer: We decoupled traditional routing into a Multi-Constraint A* Graph Engine with a Tripartite Evaluation Metric : 1) <i>Feasibility</i> (Boolean Pass/Fail); 2) <i>Quality Score (0–100)</i> (comfort, slope, surface); and 3) <i>Resilience Score (0–100)</i> (graph redundancy & Single Points of Failure count).	
Q3: What does 'Tri-State Logic (UNKNOWN != FALSE)' mean and why does it matter?	[Data Integrity & Safety]
Answer: In accessibility, assuming missing data means 'accessible' strands wheelchair users. Assuming missing data means 'inaccessible' excludes usable venues. We enforce UNKNOWN != FALSE : unverified links trigger a fallback probe, request AI photo verification, or offer alternate certified routes with zero false assumptions.	
Q4: How does real-time outage rerouting work when a lift or ramp breaks?	[Real-Time Architecture]
Answer: When an IoT sensor or venue operator flags a lift outage (e.g. Gate 3 Lift down), the graph edge weight is instantly set to infinity (INVALID). The engine executes an instant dynamic sub-graph re-evaluation, rerouting the user through Gate 5 ramp without needing to restart their entire journey.	
Q5: How does the system adapt to different types of disabilities?	[Personalization Engine]
Answer: Users set explicit profile constraints: Motor/Wheelchair filters out steps and >1:12 slopes; Visual prioritizes tactile paving and audible cues; Sensory/Autism avoids high-decibel corridors; and Elderly minimizes total walking distance and stair climbing.	

Q6: What is the exact role of Gemini AI in your project?	[AI & Computer Vision]
Answer: Gemini serves two core functions: 1) Multimodal Vision: Users snap photos of an entrance or curb, and Gemini identifies ramp presence, slope gradient, and obstacles; 2) Multilingual Voice Assistant: Generates natural audio directions in 5+ Indic languages (Hindi, Tamil, Telugu, etc.) for non-English speakers.	
Q7: Where does the initial spatial data come from? Isn't mapping hard?	[Data Pipeline & Scalability]
Answer: We use a 3-tier data aggregation model: 1) OpenStreetMap (OSM) spatial seed + Government Sugamya Bharat data; 2) B2B Venue Operator Portal where stadiums/airports upload CAD/indoor layouts; 3) Crowd-verified AI photo scans validated by community contributors.	
Q8: Can your graph routing scale to millions of routes in Indian cities?	[Scalability & Performance]
Answer: Yes. We use PostGIS spatial bounding-box indexing to prune the search space to relevant local clusters. Graph pathfinding runs on pre-filtered adjacency lists in memory with sub-50ms execution latency, easily scaling via stateless Next.js server actions.	
Q9: What is your Business & Revenue Model? Who pays for this?	[Business Model & Monetization]
Answer: 1) B2G / Smart Cities: Government licenses for city-wide accessibility audits and gap heatmaps; 2) B2B SaaS: Annual compliance and indoor navigation licenses for sports stadiums, hotels, and universities; 3) B2B2C API Layer: Data licensing to MakeMyTrip, IRCTC, and Uber; 4) Free for Divyangjan.	
Q10: How does this align with Government of India policies?	[National Policy Alignment]
Answer: SugamSetu directly accelerates the Sugamya Bharat Abhiyan (Accessible India Campaign) , enforces compliance with the Rights of Persons with Disabilities (RPwD) Act 2016 , and enhances India's readiness for hosting international Para-Sports events.	
Q11: What happens if the traveler loses cellular/internet connectivity?	[Offline & Resilience]
Answer: The journey manifest (nodes, emergency contacts, offline floor maps) is pre-cached locally on the client device using Progressive Web App (PWA) Service Workers and IndexedDB, ensuring uninterrupted guidance even inside basements or stadium tunnels.	
Q12: How do you prevent spam or false crowdsourced accessibility data?	[Data Security & Trust]
Answer: Crowdsourced reports require an attached photo. Our Gemini Vision API cross-references the photo for visual evidence of ramps/stairs, verifies EXIF geolocation metadata against the user's GPS, and applies a reputation-weighted consensus algorithm before updating the live graph.	
Q13: Why did you choose Next.js 16 and PostgreSQL over other stacks?	[Architecture Decision]
Answer: Next.js 16 App Router provides unified server/client execution, dynamic API routes, and instant SEO/PWA capabilities. PostgreSQL + PostGIS provides battle-tested ACID spatial topology querying that NoSQL databases cannot handle efficiently.	
Q14: What are your immediate next steps / roadmap after SIH?	[Future Roadmap]
Answer: 1) Pilot deployment at 3 major metro stations and 2 sports stadiums; 2) Integration with IRCTC for direct accessible train coach booking; 3) WebXR-based AR Indoor Wayfinding for real-time visual camera guidance.	
Q15: What is the single biggest USP you want the jury to remember?	[Closing Pitch Punchline]
Answer: 'Accessibility is not a point—it is a chain.' If even one link breaks, the person is stranded. SugamSetu is India's first platform that guarantees 100% unbroken journey continuity.	